

Technology News

Robot to replace manpower

Japan has found a solution for labour shortage. Its new humanoid robot—the HRP-4—is a lean, light robot that can balance itself with ease and stand on one leg, track faces and objects, and respond to voice commands.

Looking like a 'slim athlete,' the HRP-4 weighs only 39 kg (86 pounds) including battery. That's about 4 kg less than HRP-4C, which made its appearance last year at a bridal fashion show in Osaka, as well as *Japan Fashion Week* in Tokyo.

HRP-4 stands nearly 1.5 metres tall and has 34 moving joints, with seven in each arm, as well as fingers than can move more precisely than earlier HRP models. Each arm can take load of about 45 gm (1 pound). All joint motors are less than 80 watts for design safety. A compact notebook computer can be installed in the back to increase onboard data processing.

The robot has been developed by bridge builder Kawada Industries and the National Institute of Advanced



HRP-4 humanoid robot looks like a slim athlete

Industrial Science and Technology (AIST).

Seaswarm robot to clean up oil spills

Using a cutting-edge nanotechnology, researchers at Massachusetts Institute of Technology (MIT) have created a

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Unlike traditional skimmers, Seaswarm is based on a system of small, autonomous units that behave like a swarm and 'digest' the oil locally while working round the clock without human intervention. Using swarm behaviour, the units use wireless



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robotic prototype that could autonomously navigate the surface of oceans to collect surface oil and process it on site. The system, called Seaswarm, is a fleet of vehicles that may make cleaning up future oil spills both less expensive and more efficient than current skimming methods.

The Seaswarm robot uses a conveyor belt covered with a thin nanowire mesh to absorb oil. The fabric, developed by MIT's visiting associate professor Francesco Stellacci, can absorb up to twenty times its own weight in oil while repelling water. By heating up the material, the oil can be removed and burnt locally and the nanofabric reused.

The robot, which is about 5 metres (16 feet) long and 2 metres (7 feet) wide, uses two square metres of solar panels for self-propulsion. With just 100 watts (the equivalent of one household light bulb), it could poten-

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Future handhelds to offer advice

In the future, your handheld device could offer you advice that today only a friend or relative could give. So future handhelds won't be just devices but assistants or even companions. They could combine already common geographic location technology with data from microphones, cameras, heart and body monitors and even brain scans.

"Imagine a device that uses a variety of sensory modalities to determine what you are doing at any instant, from being asleep in your bed to being out